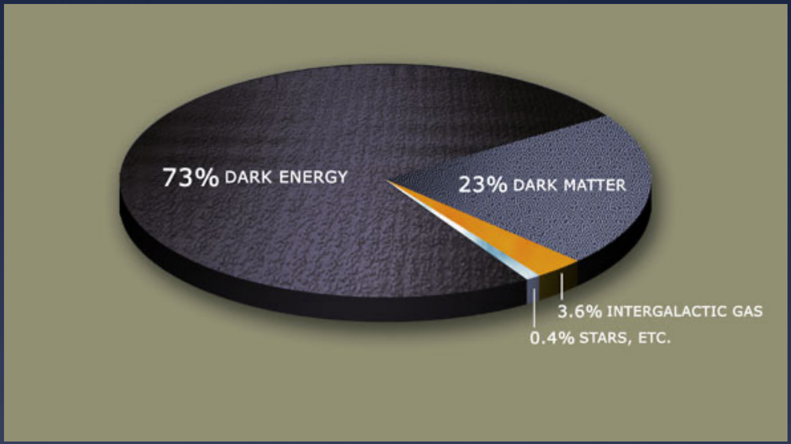
Dark Matter

By
Arun Mathew

Department of physics
Mar Thoma college
Thiruvalla
Affiliated to MG university, kottayam

What is Dark matter

- ★ Unseen matter which comprises 23% of the total mass-energy density of our universe.



- ★ Abundance of light elements and microwave background radiation suggest dark matter is non baryonic.

- ★ Electromagnetically neutral matter, there presence can only be detected through their gravitational interaction with ordinary matter.

- ★ They interact with each other through a intermediate gauge boson.

Evidence

- they are observed by its gravitational effect on luminous matter

1. Cosmological scale

$$ds^2 = c^2dt^2 - R^2(t)\left\{\frac{dr}{1-kr^2} + r(d\theta^2 + \sin^2\theta d\phi^2)\right\}$$

Friedman - Roberson - walkers metric

Where R(t) cosmic scale factor, k- curvature parameter

★ Field equation can be rewritten down as

$$\Omega = 1 + \frac{kc^2}{H^2R^2}$$

Where $\Omega=\rho/\rho_C$ and $\rho_C=3H^2/8\pi G$

★ Cosmic density parameter

$$\Omega = \Omega_M + \Omega_\Lambda \cong 1$$

$$\Omega_\Lambda = 0.73$$

$$\Omega_B = 0.04 - 0.05$$

$$\Omega \le 0.30$$

2. Galactic cluster scale

- Mass is measured by observing the mass to light ratio ϒ .
- Also by looking at the mean square velocity of the individual galaxies of the cluster.

$$M = \frac{\langle v^2 \rangle R_M}{G}$$

$$T << \mathrm{h}^2/mk_B \left(N/V\right)^{2/3}$$

The mass estimated this way shows a large deviation from the data obtained From the luminous analysis of the same cluster.

It is observed that the density of the cluster goes like

$$\rho(r) \approx \frac{1}{r^2}$$

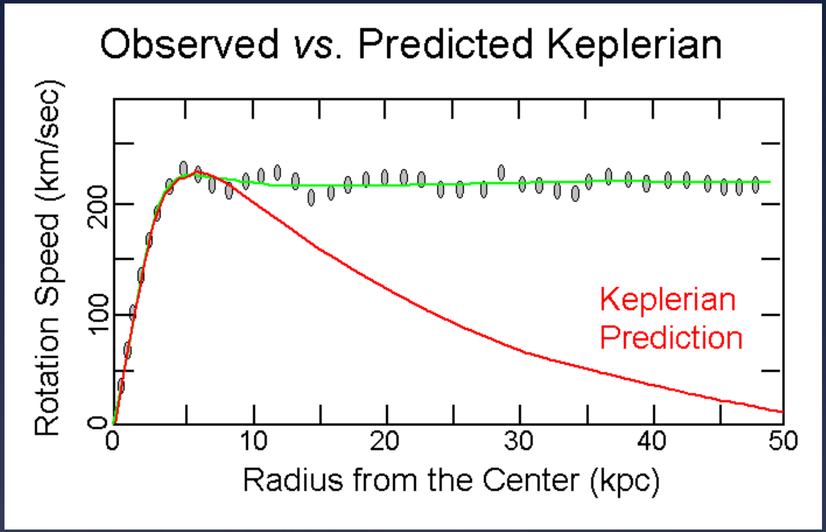
(singular isothermal spherical halo)

$$\Omega_M \leq 5.9\Omega_B$$

3. Galactic scale

Mass can be estimated by measuring the mass to light ratio ϒ of the galaxy

Total luminosity of universe is produced by the galaxy in it. we obtain this from a empirical form called luminosity function.



$$\Phi(L) = \frac{\Phi_*}{L_*}\left(\frac{L}{L_*}\right)^{\alpha} \exp^{-L/L_*}$$

Where $\Phi_* \cong 0.41 \times 10^{-3}\ \mathrm{Mpc}^{-3}$, $L_* \cong 2.53 \times 10^{10}\ L_\odot$ and $\alpha = -1.25$

Table 1: Contribution to Ω due to various galaxy populations are listed

Type of Galaxy X	$\Upsilon_{*,X}$	$\mathcal{F}_X$	$\mathcal{L}_X$ $(L_\odot Mpc^{-3})$	$\rho_{*,X}$ $(M_\odot Mpc^{-3})$	$\Omega_{*,X}$
Ellipticals E	6.5	0.11	1.54×10^7	1.00×10^8	0.73×10^{-3}
Lenticular $S0$	5.0	0.21	2.94×10^7	1.47×10^8	1.08×10^{-3}
Spirals Sa	3.0	0.28	3.92×10^7	1.17×10^8	0.87×10^{-3}
Spirals Sb	2.0	0.29	4.06×10^7	0.81×10^8	0.59×10^{-3}
Spirals Sc	1.0	0.05	0.70×10^7	0.07×10^8	0.05×10^{-3}
Irregulars	1.0	0.06	0.84×10^7	0.08×10^8	0.06×10^{-3}

Where

$\Upsilon_{*,X}$ - Luminous mass for galaxy type X.

F_X - fraction of luminosity for galaxy type X.

L_X - mean luminosity density for galaxy type X .

$\rho_{*,X}$ -related cosmic density for galaxy type X.

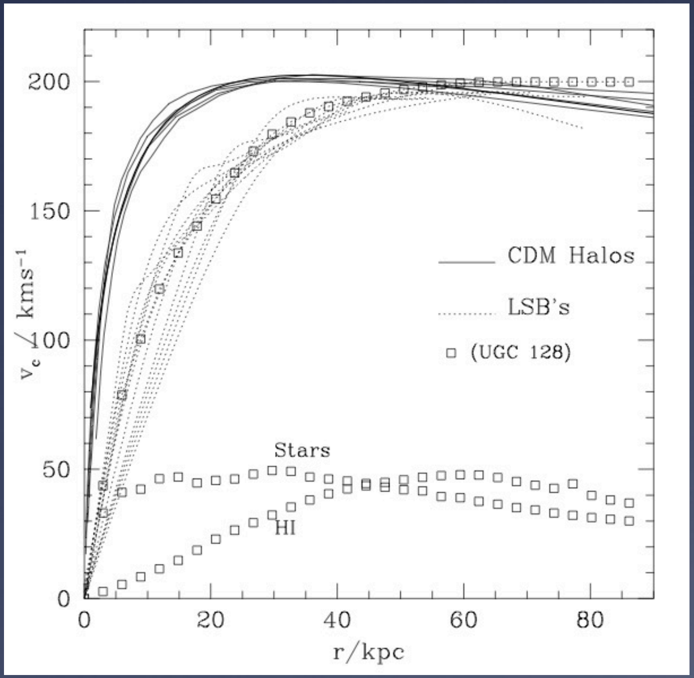
$$\Omega_* = \sum_X \Omega_{*,X} \cong 0.004$$

Candidates

They are many dark matter particles suggested

- Axions
- Sterile Neutrinos
- Higgs field
- Neutrilino
- Scalar field dark matter (SFDM)

  - This model gives good agreement both in cosmological and galactic scale.
  - It was suggested as an alternative to cold dark matter particle which fails to reproduce the rotational curve of low surface brightness galaxies



The N-body simulations such as Kra, NFW, Moore and Iso models suggest a cuspy density in the inner radius which contradict with observed flat core. This gives them the properties of self interaction and upgrade to warm. Even the warm dark matter model shows inconsistency with observations. The new scalar field model solve most of these issues.

Theory of Scalar Field Dark Matter

We start with an action

$$S = \int d^4x\sqrt{g}\left(\frac{-R}{k_0} + (\nabla\phi)^2 + V(\phi)\right)$$

The corresponding Field equations are

$$\phi^{;\mu}_{;\mu} - \frac{1}{4}\frac{dV}{d\phi} = 0$$

$$R_{\mu\nu} = k\left[2\phi_{,\mu}\phi_{,\nu} + \frac{1}{2}g_{\mu\nu}V(\phi)\right]$$

We assume halo to be symmetric with respect to the rotation axis. we also supposed that the dragging effect to be very small since the rotation motion of galaxy does not affect the motion of the test particles in it. In that case Papapetrou – line elements is the suitable metric

$$ds^2 = -e^{2\psi}(dt + \omega d\phi)^2 + e^{-2\psi}\left[e^{2\gamma}(d\rho^2 + dz^2) + \mu^2 d\phi^2\right]$$

Lagrangian for test particle travelling on the static space-time($\omega = 0$) is

$$2L = -e^{2\psi}\dot{t} + e^{-2\psi}\left[e^{2\gamma}(\dot{\rho}^2 + \dot{z}^2) + \mu^2\dot{\phi}^2\right]$$

From this constant of motions are obtained, they are E and L_φ.

$$E = e^{\Psi}\left(\frac{\mu'/\mu - \psi'}{\mu'/\mu - 2\psi'}\right)^{1/2}$$

$$L_\phi = e^{-\psi}\mu\left(\frac{\psi'}{\mu'/\mu - 2\psi'}\right)^{1/2}$$

The expression of tangential velocity of the test particle can be obtain from the metric

$$ds^2 = -e^{2\psi}dt^2 + e^{-2\psi}\mu^2 d\phi^2 + e^{-2(\psi-\gamma)}(d\rho^2 + dz^2)$$

In term of proper time, dτ²= -ds²

$$d\tau^2 = e^{2\psi}dt^2 \left\{1 - e^{-4\psi}\mu^2\left(\frac{d\phi}{dt}\right)^2 - e^{-2(2\psi-\gamma)}\left(\left(\frac{d\rho}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2\right)\right\}$$

$$1 = e^{2\psi}u^0\left[1 - v^2\right]$$

where u^0 =dt/dτ is the time component of four Velocity and v is 3- velocity of the test particle.

The 3-velocity is given by

$$v^2 = e^{-4\psi}\mu^2\left(\frac{d\phi}{dt}\right)^2 + e^{-2(2\psi-\gamma)}\left(\left(\frac{d\rho}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2\right)$$

$$v^2 = v^{(\phi)^2} + v^{(\rho)^2} + v^{(z)^2}$$

$$v^{(\phi)} = \left(\frac{\psi'}{\mu'/\mu - \psi'}\right)^{1/2}$$

We are considering the situation where tangential Velocity for circular trajectories is a constant i.e,

$$v^{(\phi)} = v_c^{(\phi)}$$

We get $$e^{\psi} = \left(\frac{\mu}{\mu_0}\right)^{l}$$

The metric is reduced to

$$ds^2 = -\left(\frac{\mu}{\mu_0}\right)^{2l} dt^2 + \left(\frac{\mu}{\mu_0}\right)^{-2l} \left[e^{2\gamma} d\rho^2 + \mu^2 d\phi^2\right]$$

Using the above equation in geodesic equation

$$\frac{d^2 x^{\mu}}{d\tau^2} + \Gamma^{\mu}_{\nu\lambda} \frac{dx^{\nu}}{d\tau} \frac{dx^{\lambda}}{d\tau} = 0$$

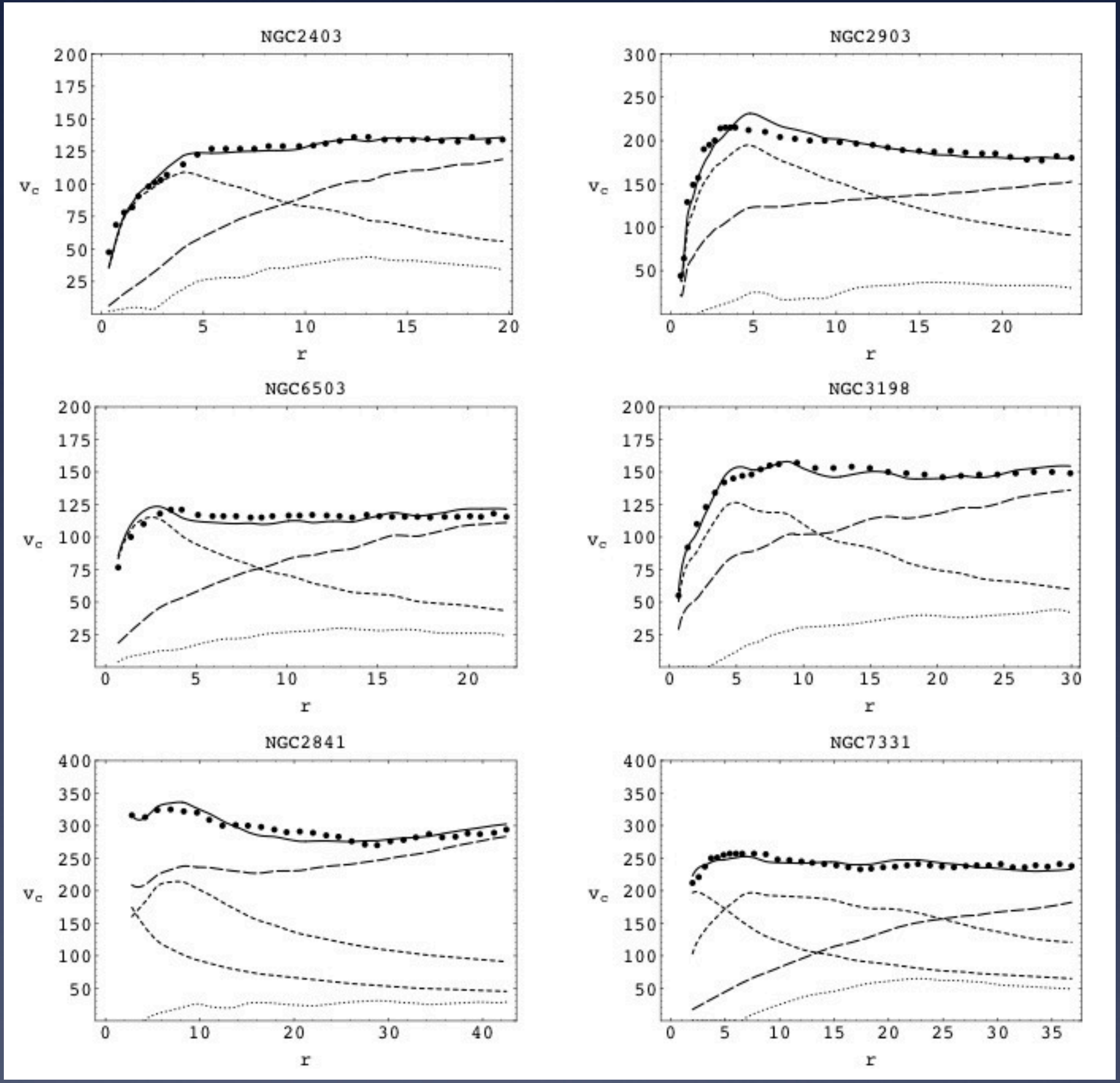
We obtain

$$v_C^2 = \frac{GM_L(r)}{r}\left(1 + f_0(r^2 + b^2)\right) + v_{gas}^2$$

Where f_0 and b are adjustable constants

Results

Curves of v_C (km/s) vs r (kpc) for the observational data (dots) and the tits from the obtained equation (solid). Also shown are the individual contributions from luminous matter (short dashed), the scalar eld (long dashed) and gas (dotted). In the case of NGC 2841 and NGC 7331, the lower short dashed curve represents the contribution from the bulge of these galaxies.



Conclusion

Many independent observation corn the existence of dark matter. It is observed that 23% of our universe contain non-baryonic dark matter. But we still do not know of what non-baryonic particles the dark matter made of.

So far the direct information of dark matter comes solely from astronomical observations, we have not found them on earth, in colliders.

Next generation experimental setup like Ice Cube, high energy neutrino telescope is our aspiration to know what this massing particles really are.

We have shown that a scalar field together with Einstein's equation suggest a cold radially pressured field would solve the difficulty.

The next question we would probably ask is whether this field is quantizable? or is the quantum particles due to this field that we should look for in colliders?

Hopefully the problem of both dark matter and dark energy will soon be resolved. It is important to know what consist of 95% of our neighborhood.

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